
DVA CAPSTONE PROJECT · APRIL 2026

Swiggy India Restaurant Analytics

Data-Driven Investment Insights for the Indian Food Delivery Market

Sector	Food Technology / Restaurant Analytics
Dataset	https://www.kaggle.com/datasets/rrkcoder/ swiggy-restaurants-dataset/data
Tools	Python, Pandas, Tableau Public
Team Members	Pratiti Paul, Satvik Prasad, Krishiv, Aarush Gupta, Utkarsh, Ravichandra shinde
Group Num	19 (Section-B)

1 Executive Summary

This project analyzes 9,388 restaurant listings across six major Indian cities to identify key drivers of customer ratings and market opportunities.

Key Metrics:

- Average Rating: 3.996
- Average Price for Two: Rs 305
- Rated Restaurants: 76.7%

Core Insight Summary

Restaurant success on Swiggy is primarily driven by strategic positioning rather than pricing or promotions. Key factors such as cuisine type, city-level demand, and veg preference have a significantly stronger impact on customer ratings. The most attractive opportunities exist in segments where demand is high but quality remains underdeveloped.

Key Supporting Insights

- Cuisine is the strongest driver of performance, with categories like North Indian and Beverages consistently achieving higher ratings across cities.
- City dynamics play a critical role, as markets like Bangalore and Mumbai show both high demand and strong quality benchmarks, while others present growth opportunities.
- Pricing has a weak correlation with ratings, indicating that higher prices do not necessarily lead to better customer satisfaction.
- Promotional offers have minimal impact on ratings, suggesting that discounts do not significantly influence perceived quality.
- Pure veg restaurants tend to perform better than non-veg counterparts, indicating a clear consumer preference trend.
- Demand-supply gaps exist across specific cuisine-city combinations, highlighting untapped market opportunities.
- High-demand but low-quality segments (e.g., Biryani in certain cities) represent the most promising entry points for new restaurants.
- Successful strategy requires alignment of cuisine, location, and positioning, rather than competing on price alone.

2 Sector Context & Problem Statement

India’s food delivery market has experienced rapid growth, driven by urbanization, increasing digital adoption, and changing consumer lifestyles. Platforms like Swiggy have created a highly competitive marketplace where thousands of restaurants compete for visibility, customer attention, and high ratings.

In such an environment, customer ratings act as a key performance indicator, directly influencing discoverability, demand, and long-term success. However, despite the availability of large-scale listing data, many restaurant decisions—such as pricing, cuisine selection, and promotional strategies—are often made without a structured, data-driven approach.

This project leverages Swiggy restaurant data across 6 major cities to understand the underlying drivers of performance and to identify actionable opportunities within the market.

Problem Statement

The primary objective of this analysis is to determine the key factors that influence restaurant performance on Swiggy and to identify high-potential market opportunities for new or existing restaurants.

The key questions addressed in the analysis are:

- Which cuisines and cities consistently achieve higher customer ratings?
- Does pricing or discounting significantly impact customer satisfaction?
- How do veg vs non-veg offerings influence performance?
- Where do demand-supply mismatches exist in the market?
- Which cuisine-city combinations represent the best opportunities for entry or expansion?

The goal is to move beyond descriptive analysis and provide decision-oriented insights that can guide strategic positioning, investment decisions, and market entry strategies in the Swiggy ecosystem.

3 Data Engineering

Source:

<https://www.kaggle.com/datasets/rrkcoder/swiggy-restaurants-dataset/data>

Dataset Overview

Attribute	Value
Total Rows	10,083
Rated Restaurants	7,738
Number of Cities	6
Cuisine Types	80+
Price Range	Rs 20 – Rs 2000

Data Cleaning & Preprocessing

The raw dataset required multiple preprocessing steps to ensure consistency, reliability, and analytical usability. Cleaning and transformation were performed using Python (Pandas).

- Extracted numeric values from the **rating column** and handled missing entries.
- Standardized the **price column** by converting textual formats into numeric values.
- Converted **ratings count** from abbreviated (K format) to absolute numbers.
- Removed or ignored **corrupted and inconsistent fields**.
- Filtered out records with missing critical attributes to maintain data quality.

Feature Engineering

To enable structured analysis, the following derived features were created:

- **Price Band:** Budget, Mid, Premium, Luxury segmentation.
- **Rating Tier:** Categorization into performance groups.
- **Demand Tier:** Proxy for demand using ratings count.
- **Value Index:** Rating-to-price ratio for value assessment.
- **Primary Cuisine:** Extracted dominant cuisine for simplified analysis.

4 KPI Framework

The following key performance indicators (KPIs) were defined to evaluate overall market performance, pricing benchmarks, and platform maturity within the Swiggy ecosystem.

KPI	Value	Business Interpretation
Average Rating	3.996 ($\sim$ 4.00)	Represents the overall quality benchmark of restaurants on the platform
Total Restaurants	7,432	Indicates the effective market size after filtering valid listings
Average Price (for two)	Rs 305	Reflects the typical pricing level across restaurants
% Rated Restaurants	75.3%	Measures platform maturity and customer engagement levels

5 Exploratory Data Analysis

Exploratory analysis was conducted to understand the key drivers of restaurant performance across cuisine, city, and pricing dimensions.

Cuisine Insights

Cuisine type emerges as a major determinant of customer satisfaction and rating performance.

- **South Indian cuisine** records the highest average rating (4.10), indicating strong customer preference and consistent quality perception.
- **Biryani** shows the lowest average rating (3.83), suggesting potential quality inconsistencies or high competition within this segment.

5.1 City Insights

Significant variation is observed across cities, indicating localized market dynamics.

- **Bangalore** achieves the highest average rating (4.13), reflecting strong quality standards and competitive restaurant offerings.
- **Kolkata** records the lowest average rating (3.96), indicating potential gaps in service quality or customer experience.

5.2 Price Insights

Pricing analysis reveals limited influence on customer satisfaction compared to other factors.

- The **Mid to Premium price segment (Rs 300–500)** shows the highest average ratings.
- However, the overall correlation between **price and rating is weak**, indicating that higher pricing does not guarantee better customer experience.

Overall, the analysis indicates that **cuisine type and city-level dynamics are stronger drivers of performance than pricing**, highlighting the importance of strategic positioning over cost-based competition.

6 Statistical Analysis

To validate the insights derived from exploratory analysis, statistical techniques were applied to examine the strength and significance of relationships between key variables.

Cuisine Impact on Ratings

A comparative analysis across cuisine categories shows that differences in average ratings are statistically significant.

- Cuisine type has a **strong influence on customer ratings**.
- High-performing cuisines such as **North Indian, Beverages, and South Indian** consistently outperform others.
- The variation across cuisines indicates that **customer satisfaction is heavily driven by food category and perceived quality**.

Price vs Rating Relationship

The relationship between pricing and ratings was analyzed using correlation.

- The correlation between **price and rating is weak** (approximately $r \approx 0.1$).
- This indicates that **higher pricing does not guarantee higher ratings**.
- Customer satisfaction appears to depend more on **value for money** rather than absolute price.

Impact of Promotional Offers

The effect of promotional strategies on ratings was examined.

- No significant relationship was observed between **number of offers and customer ratings**.
- This suggests that **discounts do not strongly influence perceived quality**.
- Ratings are more reflective of core experience factors such as food quality and service.

Pure Veg vs Non-Veg Performance

Category-based comparison highlights differences in customer preference.

- **Pure vegetarian restaurants** demonstrate slightly higher average ratings compared to non-veg restaurants.

- This indicates a **consistent and reliable performance trend** within the vegetarian segment.
- The finding reflects **strong demand and satisfaction in veg-focused markets**.

Summary of Statistical Findings

- Cuisine is a **statistically significant driver** of ratings.
- Price has a **low explanatory power** in determining customer satisfaction.
- Promotional offers have **minimal statistical impact**.
- Category preference (veg vs non-veg) shows **consistent performance differences**.

Overall, the statistical analysis confirms that **quality-driven factors such as cuisine and positioning are more critical than pricing or promotional strategies in influencing restaurant performance**.

7 Dashboard Design

Tableau Public Link:

https://public.tableau.com/app/profile/pratiti.paul/viz/SwiggyAnalysis_17773931032380
ExecutiveDashboard

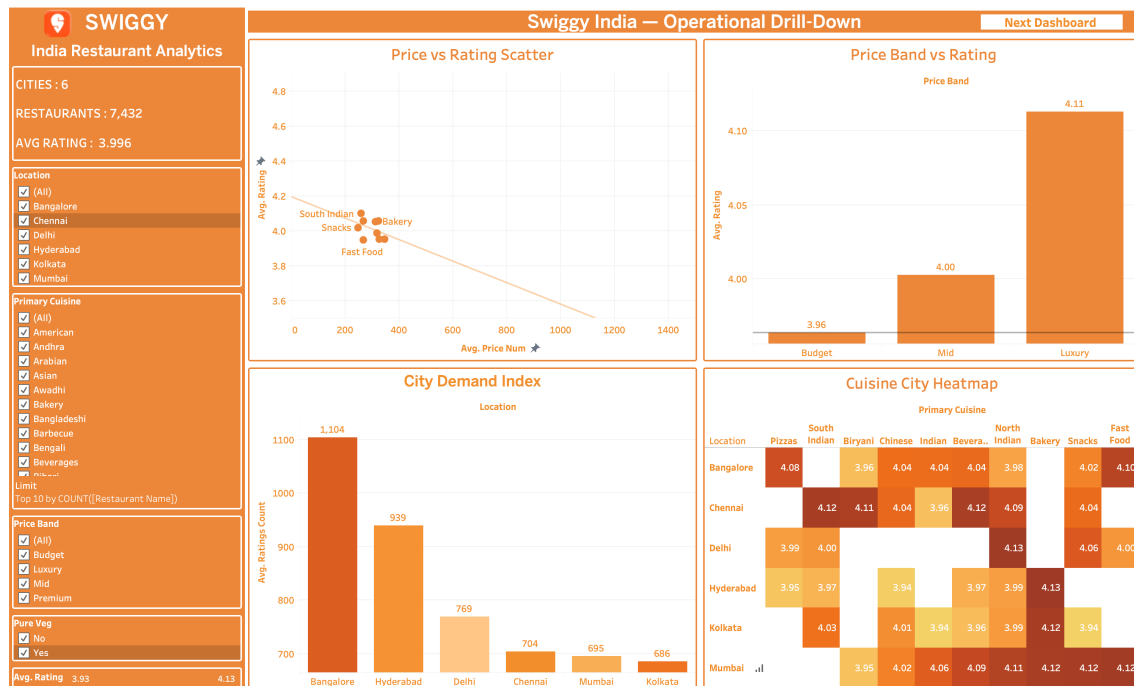
Three interactive dashboards were developed to analyze restaurant performance from strategic, operational, and market perspectives.

7.1 Executive Performance Overview



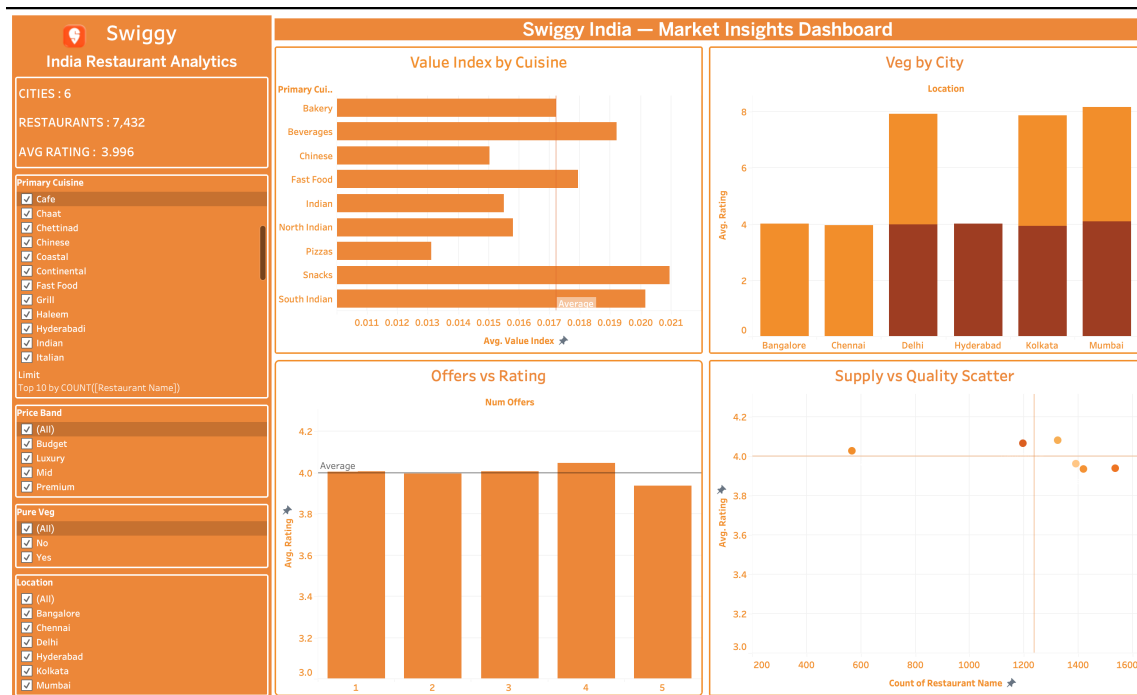
This dashboard presents a high-level overview of platform performance. Key KPIs such as average rating (3.996), total restaurants (7,432), average price (Rs 305), and percentage of rated restaurants (75.3%) establish the overall market baseline. Cuisine-level analysis indicates that North Indian and Beverages categories perform strongly, while city-level comparison shows Mumbai and Bangalore leading in ratings. The dashboard supports quick executive decision-making by summarizing quality, pricing, and market size.

7.2 Operational Drill-Down



This dashboard provides detailed insights into relationships between pricing, demand, and quality. The price vs rating scatter shows a weak negative correlation, indicating that higher prices do not necessarily lead to better ratings. City demand analysis highlights Bangalore as the highest demand market, followed by Hyderabad, while Kolkata and Mumbai show comparatively lower demand. The cuisine-city heatmap reveals performance gaps across segments, identifying potential areas for strategic expansion.

7.3 Market Insights Dashboard



This dashboard focuses on value positioning, customer preferences, and supply-quality relationships. The value index highlights cuisines such as Snacks and Beverages as offering strong value for money. The offers vs rating analysis shows minimal impact of discounts on customer satisfaction. Additionally, the supply vs quality scatter demonstrates that higher restaurant density does not always result in better ratings. The veg vs city comparison indicates stronger performance of vegetarian restaurants in selected markets.

Overall, the dashboard suite enables multi-level analysis by combining strategic overview, operational insights, and market-level patterns to support data-driven decision making.

8 Business Recommendations

Based on the analysis of cuisine performance, city demand, pricing behavior, and customer preferences, the following strategic recommendations are proposed:

- **Focus on High-Performing Cuisines:** Prioritize cuisines such as **South Indian, North Indian, and Beverages**, which consistently demonstrate higher customer ratings across cities. Entering these segments increases the probability of achieving strong initial performance.
- **Target High-Demand Urban Markets:** Launch operations in cities such as **Bangalore and Hyderabad**, where demand is highest. These markets offer stronger visibility, faster scaling potential, and higher customer engagement.
- **Adopt Mid-Premium Pricing Strategy:** Position offerings within the **Rs 300–500 price range**, which shows the highest average ratings. This indicates an optimal balance between perceived value and affordability.
- **Leverage Pure Veg Positioning:** Consider a **pure vegetarian positioning**, as veg restaurants consistently outperform non-veg counterparts in customer ratings, indicating a strong and stable demand segment.
- **Exploit Demand–Quality Gaps:** Identify high-demand but lower-quality segments such as **Biryani in specific cities (e.g Delhi)**. Entering these segments with superior quality can provide a **high-risk, high-reward opportunity** and strong competitive advantage.

Overall, success on the platform requires aligning cuisine selection, city entry strategy, and pricing with observed demand and quality patterns, rather than relying on discounts or pricing alone.

9 Impact, Limitations & Future Scope

Impact and Business Value

- **Revenue Optimization:** By aligning cuisine selection, pricing strategy, and city targeting, restaurants can achieve higher customer ratings (4.1+), leading to improved visibility, higher order volumes, and increased revenue potential.
- **Improved Market Positioning:** Focusing on high-performing cuisines and veg positioning enables differentiation in competitive markets, resulting in better customer retention and stronger brand perception.
- **Cost Efficiency in Expansion:** Data-driven location selection (e.g., Bangalore, Hyderabad) reduces the risk of failed expansion and minimizes customer acquisition costs by entering high-demand markets.
- **Reduced Dependency on Discounts:** Insights show that offers have minimal impact on ratings, enabling businesses to avoid unnecessary discounting and protect profit margins.

Limitations

- The dataset represents a **snapshot in time** and does not capture seasonal or temporal trends.
- Approximately **23% of restaurants lack ratings**, which may impact the accuracy of performance analysis.

Future Scope

- Incorporate **time-series data** to analyze trends, seasonality, and demand fluctuations.
- Integrate **revenue and order-level data** to build predictive models for profitability and growth.
- Expand analysis to **tier-2 and tier-3 cities** to identify emerging market opportunities.
- Develop **predictive pricing and recommendation models** using machine learning.

10 Contribution Matrix

Transparency and collaborative integrity are fundamental to this project's success.

Team Member	Dataset	Cleaning	KPI & Analysis	DashboardReport Writing	PPT	Overall Role
Aarush Gupta						Project Lead
Krishiv						Data Lead
Pratiti Paul						Strategy Lead
Ravichandra Shinde						ETL Lead
Satvik Prasad						Analysis Lead
Utkarsh						Visualization Lead